Chapter 9: Propagation of Light

Grade 10 Physics - Summary

1. The Principle of Rectilinear Propagation

The fundamental rule of optics is the **Principle of Rectilinear Propagation of Light**:

> *In a homogeneous, transparent medium, light always travels in straight lines.*

- **Why it matters:** This explains why we cannot see around corners without mirrors, why objects cast sharp shadows, and how optical instruments (like telescopes and microscopes) are designed.
- **Interactions:** Light will continue in a straight line forever unless it hits an object. When it hits an object, it can undergo:
  1. **Reflection** (bouncing off)
  2. **Refraction** (bending as it enters a new medium)
  3. **Absorption** (being soaked up and converted to heat)

2. Sources of Light

To see anything, light must enter our eyes. Objects are classified by how they provide that light:

- **Luminous Objects (Primary Sources):** Objects that generate and emit their own light (e.g., The Sun, a lightbulb, a candle flame).
- **Illuminated Objects (Secondary Sources / Non-luminous):** Objects that do not produce their own light. We can only see them because they *reflect* light from a luminous source into our eyes (e.g., The Moon, a mirror, this piece of paper).

3. Light Rays vs. Light Beams

Because light travels in straight lines, we use geometric lines to represent it.

- **Light Ray:** A single, straight imaginary line with an arrowhead showing the exact direction light is traveling.
- **Light Beam:** A collection or bundle of light rays traveling together.

Types of Light Beams:

1. **Cylindrical (Parallel) Beam:** All rays are parallel to each other. The width of the beam stays the same (e.g., a laser pointer).
2. **Converging Beam:** Rays travel toward a single, common meeting point called the focus (e.g., light passing through a magnifying glass).
3. **Diverging Beam:** Rays spread outward from a single starting point (e.g., light coming from a flashlight or a candle).

4. Optical Systems: Objects and Images

An **optical system** (like a mirror or a camera lens) takes light from an *object* and redirects it to form an *image*.

- **Plane Mirrors:** Form images that are the exact same shape and size as the original object.
- **Cameras / Lenses:** Bend light rays to focus them. The resulting image is usually a different size than the object (often smaller to fit on the camera's sensor/film).

5. Types of Images

Images formed by optical systems fall into two categories:

- **Real Images:** Formed when light rays *actually meet* at a point. These images are inverted (upside down) and **can** be projected onto a physical screen or sensor (like in a camera or movie theater).
- **Virtual Images:** Formed when light rays appear to come from a point, but don't actually meet there. These images are upright and **cannot** be projected onto a screen (like your reflection in a bathroom mirror).